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Biogas production enhancement using semi-aerobic pre-aeration in a hybrid bioreactor landfill

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ABSTRACT

Landfilling continues to be one of the main methods used in managing Municipal Solid Waste (MSW) worldwide, particularly in developing countries. Although in many countries national legislation aims to reduce this practice as much as possible, landfill is a necessary and unavoidable step in closing the material cycle. The need for innovative waste management techniques to improve landfill management and minimize the adverse environmental impact produced has resulted in an increasing interest in innovative systems capable of accelerating waste stabilization. Landfill bioreactors allow decomposition kinetics to be increased and post-operational phase to be shortened; in particular, hybrid bioreactors combine the benefits afforded by both aerobic and anaerobic processes. Six bioreactor simulators were used in the present study: four managed as hybrid, with an initial semi-aerobic phase and a second anaerobic phase, and two as anaerobic control bioreactors. The main goal of the first aerated phase is to reduce Volatile Fatty Acids (VFA) in order to increase pH and enhance methane production during the anaerobic phase; for this reason, air injection was stopped only when these parameters reached the optimum range for methanogenic bacteria.

Biogas and leachate were constantly monitored throughout the entire methanogenic phase with the aim of calibrating a Gompertz Model and evaluating the effects of pre-aeration on subsequent methane production. The results showed that moderate and intermittent pre-aeration produces a positive effect both on methane potential and in the kinetics of reaction.

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1. Introduction

Landfilling continues to be one of the main methods used in managing Municipal Solid Waste (MSW) worldwide, particularly in developing countries. Although in many countries national legislation aims to reduce this practice as much as possible, landfill is a necessary and unavoidable step in closing the material cycle (Cossu, 2009). The need for innovative landfill management techniques to better minimize their adverse environmental impact, has led to an increasing interest in bioreactors, capable of accelerating the decomposition of MSW and reducing the post-operational phase of a landfill (Price et al., 2003; Raga and Cossu, 2013; Xu et al., 2014).

Bioreactor landfills are characterized by use of a series of technologies including water and/or air injection, leachate recirculation, and other combinations of in situ treatments to allow adjustment of pH, redox conditions and moisture content with the aim of creating a more suitable environment for degradation processes. Moisture control is required to support metabolic processes, transport of nutrients and microorganisms (Norbu et al., 2005), while air injection speeds up degradation processes (Cossu et al., 2003; Ritzkowski and Stegmann, 2013).

In recent years, it has been demonstrated that bioreactor landfill technology promotes the decomposition and stabilization of biodegradable organic waste, through water addition, leachate recirculation or air injection (Sun et al., 2013; Erses et al., 2008; Sekman et al., 2011; Sandip et al., 2012; Xu et al., 2014; Shao et al., 2008; He et al., 2011; Raga et al., 2015). In particular, as far as the metabolic pathway is concerned, three main types of landfill bioreactors are currently considered: anaerobic, aerobic and hybrid, the latter providing for a sequence of aerobic and anaerobic conditions (EPA, 2015). Most of the bioreactor landfills discussed in literature were operated under anaerobic conditions (Price et al., 2003; Valencia et al., 2011; Vigneron et al., 2007), causing ammonia accumulation and at times a partial or complete inhibition of methane production. On the contrary, a system operated solely under aerobic conditions may increase the degradation

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kinetics of organic substances but inhibits completely methane generation and renders energy recovery impossible. Hybrid bioreactors operate under various combinations of aerobic and anaerobic conditions to achieve benefits from both (Long et al., 2009; Xu et al., 2014). The facultative landfills bioreactor is another type of Hybrid bioreactor which provides external aerobic pre-treatment of leachate prior to its recirculation in an anaerobic bioreactor, to allow both the nitrification and denitrification process to remove nitrogen compounds (De Abreu et al., 2005; US639895, 2002).

The test proposed in this study takes into consideration six landfill simulating bioreactors: four hybrid bioreactors with a first semi-aerobic phase prior to a second anaerobic phase and two anaerobic control bioreactors. During the semi-aerobic phase, air was injected into the waste mass to simulate natural convection caused by the temperature gradient between the external environment and the landfill body, with the goal of Volatile Fatty Acids (VFA) reduction and pH increase to enhance methane production in the next anaerobic phase. The effects of continuous and intermittent aeration were compared, the latter appearing beneficial for accelerating waste stabilization and reducing the acid formation, based on the outcomes of previous experiments (Sang et al., 2009). This hybrid system also provided for leachate recirculation to ensure better nutrient distribution and appropriate moisture content.

This test is based on S.An.A Landfill model (Semi-aerobic, Anaerobic, Aerated), a hybrid system featuring an initial semi-aerobic phase to enhance methane production occurring in the subsequent anaerobic step and, finally, forced aeration for the abatement of residual emissions (Repetti et al., 2013).

The change from semi-aerobic to anaerobic conditions was challenging: no aeration time was decided a priori, as this was highly dependent on the initial waste characteristics, but was selected with the aim of monitoring specific chemical parameters capable of guaranteeing optimal conditions for methanogenic bacteria.

The aim of the present study was to apply the hybrid waste management system, in order to combine the previously demonstrated advantages of leachate recirculation and aeration, with a greater methane production. In line with the mode of aeration and with air flow in each bioreactor at the moment of change from semi-aerobic to anaerobic operating conditions, an association with the characteristic parameters of methane generation (production rate, cumulative volume produced and lag phase) was observed. This analysis is of use in determining the optimum air injection method and airflow range to both stimulate methane production and increase degradation kinetics.

2. Materials and methods

2.1. Waste samples

The waste used for the experiment was the residual fraction of Municipal Solid Waste sampled after source segregation and provided by a public waste management company operating in thirteen municipalities of Livorno Province, Tuscany (Italy). A 200 kg sample was sieved with an 80 mm mesh and then analyzed by different waste fractions (Table 1); the undersieve (<80 mm fraction) was loaded into bioreactors. The waste composition was characterized by high amounts of food waste, plastic, paper and fines. Minor quantities of green waste, metals, glass, wood and textiles were detected (Table 1).

2.2. Equipment

The experiments were carried out by using six Plexiglass® (polymethyl methacrylate) columns (height 106 cm, diameter 24 cm). Each reactor was filled with the aim of reaching a density of 0.5 t/m^3, resulting in 18.4 kg MSW placed in each reactor. A 10-cm thick gravel layer (Ø 20–30 mm) was placed at the bottom and the top of each column as a drainage layer to facilitate the distribution of recirculated leachate (Fig. 1).

Table 1
Fractional composition of raw waste and undersieve (<80 mm), i.e. the fraction used to fill the reactors.

Trade class	Raw waste (%)	Undersieve <80 mm (%)
Food waste	14.0	17.4
Green waste	6.1	6.0
Paper	22.9	10.5
Plastic	18.4	12.6
Metal	5.2	6.2
Glass and inert	3.5	3.2
Wood	0.4	0.3
Textiles	8.5	5.1
Undersieve #20 mm	20.9	38.8

The upper end of reactor body was equipped with three valves providing for the introduction of air, sampling and extraction of gas as well as introduction of water and leachate recirculation. Recirculation was carried out from the top of the reactor using peristaltic pumps (Heidolph PD 5001). A leachate collection port was located at the bottom of each column for leachate extraction.

To channel air into the waste body, a vertical PVC pipe with side perforations was installed at the center of the waste layer; this system was designed to guarantee the uniform distribution of air throughout the reactor. The pumping equipment was a Prodac Air Professional pump 360 and inlet airflow was regulated by a Sho-Rate GT1135 flow meter. Biogas generated from each column was collected by a Tedlar® sampling bag connected to the upper gas port and its volume and quality were measured daily (Fig. 1).

Temperature monitoring was performed by means of Thermo Systems TS100 temperature probes installed inside the reactor and the temperature was maintained constant at 39–42 °C by means of a thermo-regulated insulation system covering all reactor lateral surfaces.

2.3. Methodology

The test was divided into three distinct main phases: Semi-aerobic, Anaerobic and Aerobic. The first phase was meant to enhance biogas production in the following anaerobic phase; the third one was functional to the complete stabilization of residual contaminants by means of aeration and flushing. In this paper, only the first two phases have been taken into consideration.

Six column reactors were filled with the same quantity of waste and managed under identical hydraulic conditions to obtain a clear comparison of results. Four reactors were Hybrid (IAa and IAb with intermittent aeration; CAa and CAb with continuous aeration); the remaining two were anaerobic (ANa and ANb) and used as control bioreactors. After filling and startup, aeration was commenced according to the scheme in Fig. 2, to simulate semi-aerobic conditions. Daily air flux was the same for all pre-aerated reactors (Fig. 2) and equal to 50 L/d at 20 °C, that in our specific case corresponds to 5 NL/d/kgTS, TS being the dry matter content (total solids) in the reactor. Intermittent aeration was turned on for twelve hours a day (4.1 L/h), while continuous aeration was provided all day long, seven days per week (2.1 L/h). Daily air flux was chosen on the basis of a series of lab-scale experiments carried out on landfill bioreactors in semi-aerobic conditions: these ranged from 0.7 NL/d/kgTS (Wu et al., 2014) to 32.0 NL/d/kgTS (Cossu et al., 2003) with average values between 4–10 NL/d/kgTS (Cossu

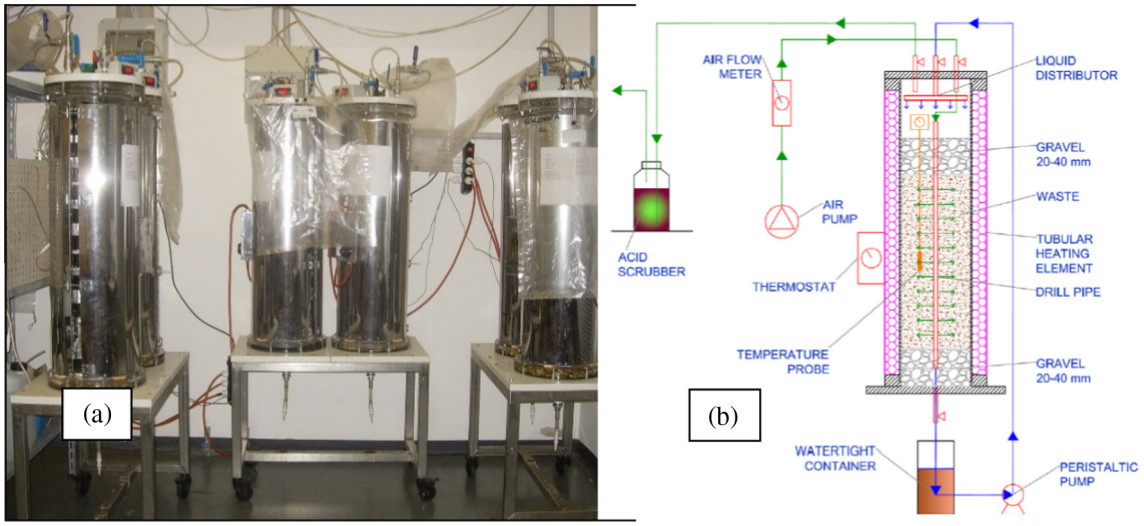


Fig. 1. Reactor setup: (a) view and (b) sketch.

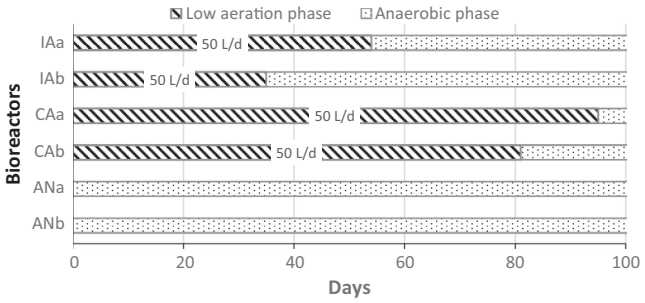


Fig. 2. Operative conditions observed over time in the different experimental columns. *IA*: intermittent aeration, *CA*: continuous aeration, *a*, *b* double series of columns. Daily aeration was the same for all pre-aerated columns (50 L/d at 20 °C and 1 atm).

Table 2
Operative parameters and range of values selected in defining the length of the pre-aeration phase.

Parameter	Values range	References
Temperature (°C)	30–40	Christensen et al. (1996)
	35–40	Yuen et al. (1995)
	35	Cossu et al. (2005)
pH	6–8	Zehnder et al. (1982)
	6.7–7.4	Lay et al. (1998)
	6.4–7.2	Chugh et al. (1998)
	6.5–8.2	Sekman et al. (2011)
	6.65–7.41	Sandip et al. (2012)
Alkalinity ($mgCaCO_3/L$)	1000–5000	Agdag and Sponza (2005)
	2000–3500	Sekman et al. (2011)
Volatile fatty acids ($mgCH_3COOH/L$)	<6000	Wang et al. (1999)

et al., 2005; Sun et al., 2013). At test start up, airflow was set at a low regime and incremented until oxygen was detectable in the off gasses (O_2 > 1–2%).

At the beginning of the test, 5 L of distilled water were introduced into each reactor to adjust moisture content to around 55–60% and guarantee leachate production; leachate recirculation was carried out daily throughout the entire test. Leachate samples (0.1–0.5 L) were collected weekly for analysis. Fresh water was injected to replace the leachate sampled for testing and to maintain a constant moisture content, fundamental for biochemical processes (Valencia et al., 2009); this represented the sole contribution to the increase of Liquid–Solid ratio (L/S).

The transition from the semi-aerobic phase to the anaerobic one was based on the observation of parameters of concern in leachate. In particular, pH steadily increased and reached values above 6 after four weeks of aeration. It was then decided to interrupt aeration in column IAb at day 35, the pH being equal to 6.25 and within the optimal range (6–8) suggested by Zehnder et al. (1982), although VFA concentration was approximately equal to 7100 $mgCH_3COOH/L$, still above the optimal value for starting methanogenesis (6000 $mgCH_3COOH/L$) recommended in Table 2. For comparison, column IAa was turned anaerobic 20 days later than IAb, the pH value being at 6.46 and VFA concentration equal to approximately 5300 $mgCH_3COOH/L$, within the optimal value in Table 2.

The transition into anaerobic conditions was postponed for columns CAa and CAb, until the pH in leachate reached values steadily above 6. Column CAb was turned anaerobic at day 81 (the pH being equal to 6.34), although VFA concentration was approximately equal to 6300 $mgCH_3COOH/L$, still above the optimal value in Table 2. For comparison, column CAa was turned anaerobic two weeks later than CAb, the pH value being at 6.46 and VFA concentration equal to approximately 5500 $mgCH_3COOH/L$, within the optimal value in Table 2 (see Table 4).

A kinetic model derived from the Gompertz Growth Equation, calibrated with biogas data, was used to mathematically simulate Methane production. This model has a sigmoid shape with a clear inflection point and its output is the cumulative volume of methane produced from the bioreactors according to Eq. (1) (Lay et al., 1998; Sandip et al., 2012).

$$BG = BGP \cdot \exp\left\{-\exp\cdot\left[\frac{R_m \cdot e}{BGP}(\lambda - t) + 1\right]\right\} \quad (1)$$

BG is the cumulative methane yield (mL/gVS), t is the cumulated time (d), BGP is the methane yield potential (mL/gVS), R_m is maximal daily methane yield (mL/gVS/d), λ is bacteria growth lag time (d) and e is a mathematical constant (=2.718). The calibration of Gompertz Equation was carried out by minimizing mean square deviation of the values measured and calculated.

2.4. Analytical methods

International standard methods were used for the analysis of solid samples, leachate and biogas.

A portable analyzer (Eco-Control LFG20) was used for periodic measurements of CO_2, CH_4 and O_2 concentrations in biogas collected in the bags.

Leachate samples were analyzed periodically for pH, VFA, alkalinity, COD, TOC, BOD_5, TKN, $N–NH_4^+$, SO_4^{2-} and Cl^-. The ratio between VFA and alkalinity was calculated to obtain the so called FOS/TAC (Voss et al., 2009). FOS/TAC is an index developed in Germany as a control parameter for anaerobic digestion processes, yielding a rapid indication as to the status of the system in relation to buffer capacity and VFA content. For this reason, FOS/TAC was monitored frequently during the first phase of the test.

Waste was thoroughly mixed before filling the reactors, in such a way that the same material was used in each test; waste samples were taken for characterization before filling the reactors. Waste characterization at the end of the anaerobic phase involved the complete extraction of the waste from each reactor and the mixing in a tank to increase the homogeneity; after this procedure, a sample of 500 g was taken and analyzed for each column. Solid samples were milled to 4 mm prior to determination of TS, VS, TKN, $N–NH_4^+$, TOC, IR_4 and IR_7. TOC on solid samples was measured by means of a TOC–VCSN Shimadzu Analyzer. Respiration Index (RI_4 and RI_7 mgO_2/gTS) was determined by means of Sapromat apparatus (H + P Labortechnik, Germany). A leaching test was carried out according to the standard UNI EN 12457-2 on waste samples (L/S was brought to 10 L/kgTS, mixing for 24 h and filtration at 0.45 μm). The eluate obtained was analyzed to evaluate the emission potential of the same contaminants considered in leachate (pH, VFA, alkalinity, COD, TOC, BOD_5, TKN, $N–NH_4^+$, SO_4^{2-} and Cl^-. All tests on solid samples were performed in duplicate.

3. Results

3.1. Solids

The results of waste characterization at the beginning of the test as well as at the end of the anaerobic phase are reported in Table 3. Waste mass decreased in 305 days from the initial value of 18.4 kg per column, to values between 14.7 and 16.9, indicating a drop between 8% and 20% in each reactor.

Total Solids (TS) content in the waste decreased from 55% to 35–45%; Volatile Solids (VS) displayed a similar trend due to the degradation processes. TKN content slightly decreased (15–18%) in all columns; nitrogen emissions occurred only via leaching and no nitrification was detected, even during the semi-aerobic phase. TOC decreased about 30% from the beginning of the test, mainly due to biogas emissions.

The leaching test on solid samples showed a significant decrease of all parameters affected by the biochemical processes taking place during the 305 test days, with the exception of ammonium, which was produced by hydrolysis from organic nitrogen and an increase was observed in the solid phase compared to initial values. Chloride emission potential was halved from the beginning to the end of the test due to fast dissolution into leachate, as reported in other similar landfill simulation experiments (Fellner et al., 2009).

Waste respiration index (RI_4) dropped significantly from the initial value in all reactors. However, except for CAb, final RI_4 values in all reactors were higher than the reference limit of 5 mgO_2/gTS indicated for waste acceptance in landfills in the German regulation (Ritzkowski et al., 2006).

Despite all analyses being performed in triplicate, the variation of analytical values was strongly influenced by heterogeneity of the solid sample, thus resulting in the lack of any evident distinction due to the diversity of management of the reactors (Table 3).

3.2. Leachate

The evolution of the main parameters monitored in leachate samples extracted from the reactors during the test is visible in Figs. 3–5.

The beneficial effect of pre-aeration on pH increase in the subsequent anaerobic phase is clearly visible. Actually, pH increase did not start until 3 weeks after aeration was stopped in IAb. Conversely, pH increase was observed just a couple of days after IAa (where aeration lasted 20 days longer than in IAb) was turned anaerobic.

The evolution of VFA in IAa and IAb is similar and it is consistent with pH results. The same behavior was observed for continuously aerated reactors.

In anaerobic reactors a steady pH increase and concomitant VFA decrease appeared after 6 months of test only.

The evolution of alkalinity was very similar in all reactors over the first 3 months of test. A clear increase trend was observed in IA reactors 2 months after aeration was stopped; the same increase pattern is visible in CA reactors from approximately 40 days after the end of aeration.

Table 3
Quality parameters of waste sample at the beginning of the test (initial waste) and at the end of the anaerobic phase (IAa, IAb, CAa, CAb, ANa, ANb): concentrations in the solid phase and in the eluate of a standard leaching test (UNI EN 12457-2).

		Initial waste	IAa	IAb	CAa	CAb	ANa	ANb
Solid phase	Waste mass (kg)	18.4	14.9	15.5	15.3	14.9	14.7	16.9
	TS (%)	55.5	39.4	35.7	47.9	44.1	61.4	41.3
	VS (%TS)	58.9	40.6	39.5	32.4	31.0	25.8	44.2
	IR_4 (mgO_2/gTS)	76.9	7.2	8.3	6.5	2.8	8.1	13.9
	IR_7 (mgO_2/gTS)	79.3	9.3	11.4	8.9	4.5	11.8	20.8
	TKN (mgN/kgTS)	9700	7596	7317	8013	8498	7683	7586
	TOC (gC/kgTS)	368	180	231	208	303	282	274
Eluate from leaching test	pH	6.2	8.5	8.4	8.3	8.3	8.1	8.3
	COD (mg/L)	14,682	4939	4791	2939	4836	2889	4834
	BOD_5 (mg/L)	7504	365	309	308	337	393	477
	Alkalinity ($mgCaCO_3/L$)	274	985	1038	783	842	587	1056
	VFA ($mgCH_3COOH/L$)	728	154	143	120	141	107	121
	TKN (mgN/L)	756	490	406	237	411	131	261
	$N–NH_4^+$ (mg/L)	78	431	265	141	236	66	98
	Organic nitrogen (mg/L)	679	59	141	96	175	65	163
	Cl^- (mg/L)	619	343	243	233	362	376	324

A behavior very similar to VFA was observed for COD and TOC also.

BOD_5/COD ratio is an important parameter that indicates the amount of biodegradable compounds still present in a liquid (Cossu et al., 2012). According to Sekman et al. (2011), BOD_5/COD ratio between 0.02 and 0.13 implies a low biodegradability, while according to Kjeldsen et al. (2002), a ratio between 0.4 and 0.8 implies high biodegradability. During the test, BOD_5/COD ratio decreases in all reactors starting couple of weeks after the beginning of methanogenic phase and reaching values under 0.1 (Fig. 4).

The results of monitoring of nitrogen compounds are reported in Fig. 5. Nitrification processes may occur in aerobic bioreactors and allow the conversion of ammonium into nitrites and nitrates. During the first semi-aerobic phase of this test however, no nitrates were found in the leachate samples and no ammonium decrease was detected, probably due to the low quantity of air injected. Ammonium nitrogen concentration accumulated in all bioreactors due to hydrolysis of organic nitrogen and in IAa, CAa, IAb, CAb reached values of almost 2000 mgN/L during the first months of test. An unexpected increase trend was observed in aerated reactors after day 150, with values up to 3000 mgN/L in IAa. Concentrations were below 1500 mgN/L in ANa and ANb due to slower ammonification processes under anaerobic conditions.

3.3. Biogas production

The evolution of cumulative methane production is reported in Fig. 6. In hybrid reactors, methane production started within the first 5–7 days under anaerobic conditions, reaching a peak after 2 weeks; in control bioreactors ANa and ANb methane production started after 5 and 6 months respectively, from the beginning of the test.

Cumulative methane produced in the whole test in reactors IAa and IAb was 75 and 102 NL/kgVS respectively; the lower value recorded for IAa is most probably due to the higher amount of organic material oxidized during the longer pre-aeration phase.

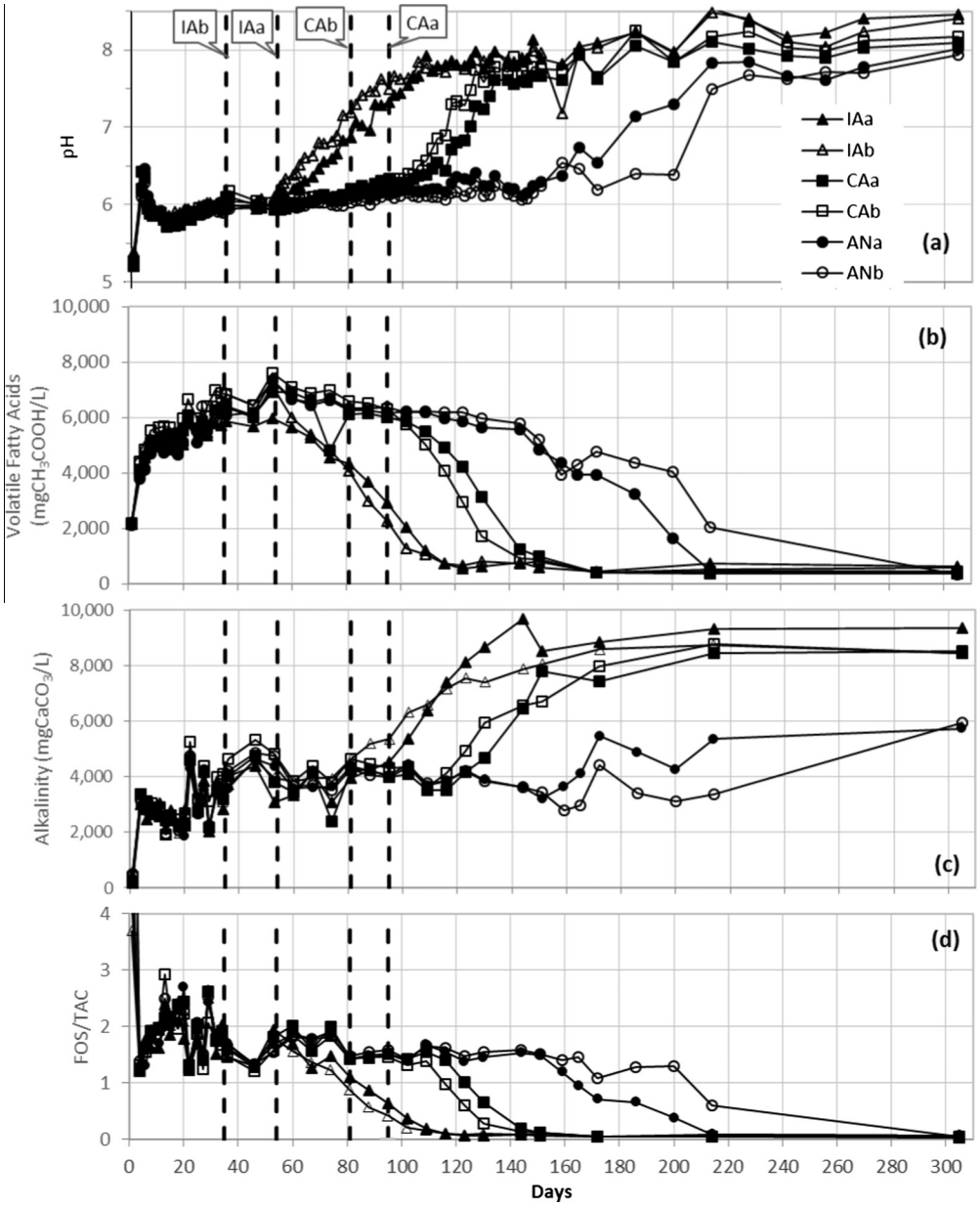


Fig. 3. Monitoring of control parameters in leachate, useful in selecting the appropriate time to turn off aeration and start the second phase of the test: (a) pH, (b) Volatile Fatty Acids (VFA), (c) Alkalinity, (d) FOS/TAC ratio. Dotted vertical lines indicate the passage form the first to the second phase of hybrid reactors.

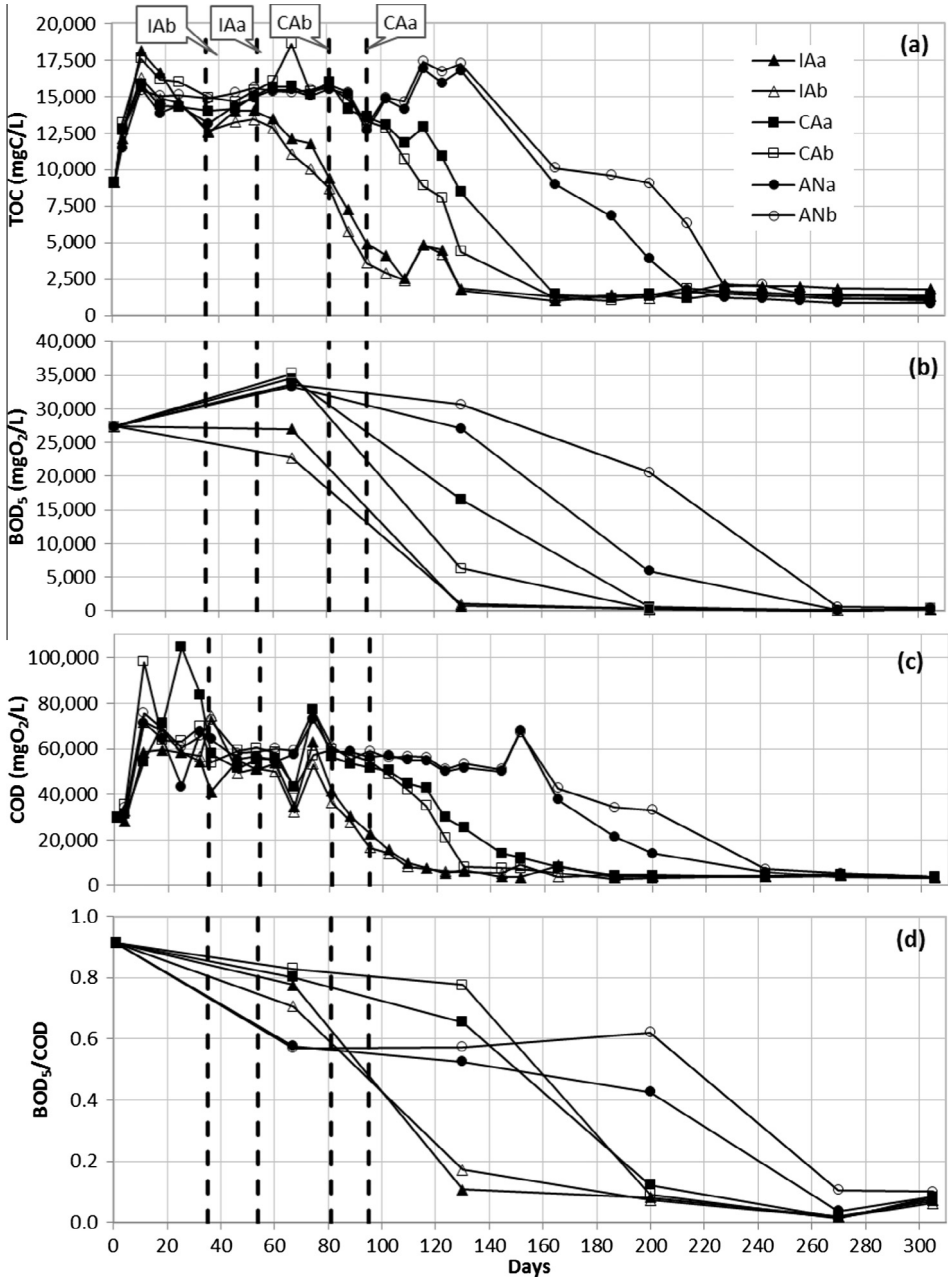


Fig. 4. Carbon indexes monitoring in leachate: (a) TOC, (b) BOD_5, (c) COD, (d) BOD_5/COD ratio. Dotted vertical lines indicate the passage form the first to the second phase of hybrid reactors.

Values between 54 and 80 NL/kgVS were obtained for continuous reactors, between 31 and 55 NL/kgVS for control bioreactors (Table 5). The highest methane production rate was recorded in intermittent reactors IAa and IAb, where a peak was observed on day 86 at 2.9 NL/d/kgVS for IAa and on day 96 at 2.8 NL/d/kgVS for IAb (Table 5).

4. Discussion

4.1. Mass balance

Carbon and nitrogen mass balances were built considering the TOC and TKN content in the solid sample at the beginning of the test and at the end of the anaerobic phase, the cumulative emissions in leachate and biogas (Table 6).

Carbon mass balance evidenced a difference between the analyzed and the calculated final content of TOC of up to 35%, with a relative error of 19%: this was probably due to waste heterogeneity rather than system losses. Liquid emissions totalled approximately 2–3% of the initial carbon content with no evident differences between pre-aerated and control reactors. On the contrary, gaseous emissions varied from 12% to 17% of initial TOC in pre-aerated reactors and only 6–8% in control ones, indicating the effectiveness of pre-aeration in methane generation enhancement, as already observed by other authors (Sekman et al., 2011; Sang et al., 2009; Erses et al., 2008). These data also confirmed that carbon emissions via leachate are generally an order of magnitude lower than those emitted in biogas (Lornage et al., 2013).

Nitrogen emissions were all associated to TKN in leachate; no N–NOx were detected during the test and denitrification was supposed to be negligible as well. Liquid emissions ranged from 12% to

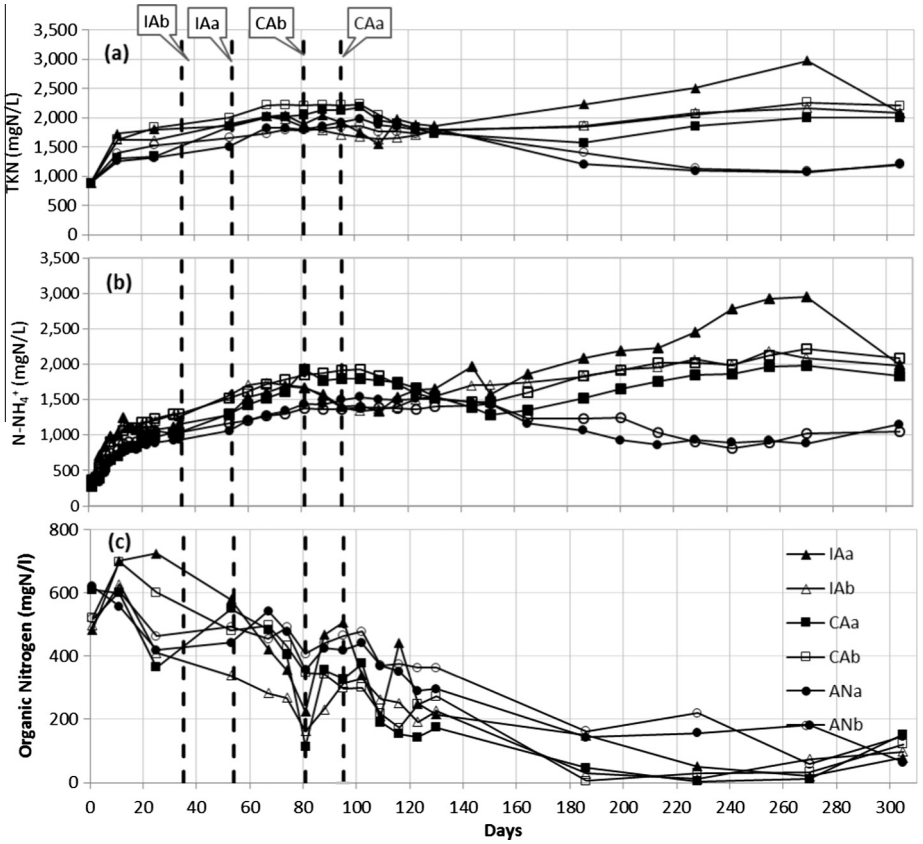


Fig. 5. Nitrogen parameters monitoring in Leachate (a) TKN, (b) $N–NH_4^+$, (c) Organic Nitrogen. Dotted vertical lines indicate the passage form the first to the second phase of hybrid reactors.

Table 4
Values of the operative parameters (listed in Table 2) monitored in the pre-aerated columns at the end of the semi-aerobic phase.

	Aeration time (d)	pH	VFA ($mgCH_3COOH/L$)	Alkalinity ($mgCaCO_3/L$)	FOS/TAC	$N–NH^{4+}$ (mg/L)	COD (mg/L)
IAa	55	6.46	5287	4163	1.27	1597	34,778
IAb	35	6.25	7144	3953	1.81	1557	51,465
CAa	95	6.46	5479	3533	1.55	1766	44,764
CAb	81	6.34	6283	4321	1.45	1918	54,344

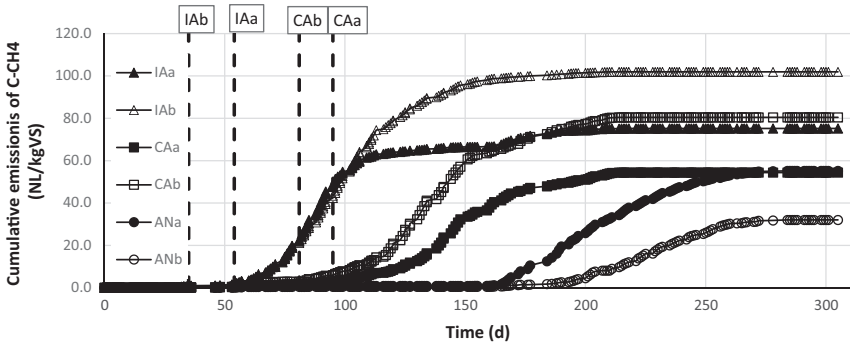


Fig. 6. Cumulated volume of methane produced in time. Dotted vertical lines indicate the passage form the first to the second phase of hybrid reactors.

15% of the initial nitrogen content. No gaseous ammonia was emitted via biogas, although pH achieved values higher than 8 at the end of the test. The difference between the analyzed and calculated final nitrogen content was approximately (−8)–5%, with a relative error of 3.8%, probably mainly due to heterogeneity of the waste samples. Dissolved nitrogen increased due to ammonification process that took place mainly in pre-aerated reactors; this was confirmed both by concentration evolution (Fig. 5) and by total nitrogen emissions, which were higher in all hybrid reactors in comparison with control ones (Table 6).

Table 5
Methane production observed at the end of the anaerobic phase in the experimental columns, according to the different aeration conditions adopted. Values of the Gompertz parameters calculated for the different columns are also reported.

	Experimental data				Gompertz model parameters			
	Total volume of injected air (NL/kgVS)	Pre-aeration time (d)	Maximum daily methane production (NL/d/kgVS)	Cumulative methane production (NL/kgVS)	BGP (NL/kgVS)	R_m (NL/d/kgVS)	λ (d)	Standard deviation
IAa	357	55	2.9	75.2	72.7	1.6	12.3	6.9
IAb	210	35	2.8	101.9	102.3	1.7	34.2	0.8
CAa	675	95	2.3	54.4	55.5	0.8	15.7	3.2
CAb	566	81	2.5	80.5	81.3	1.2	21.5	5.8
ANa	0	0	1.6	54.7	55.8	0.8	168	1.0
ANb	0	0	1.2	31.2	34.8	0.5	192	1.4

Table 6
Carbon mass balance based on TOC content detected at the beginning and at the end of the test in solid samples, TOC emissions via leachate and via biogas. Nitrogen mass balance based on TKN content detected at the beginning and at the end of the test, nitrogen emissions via leachate and via biogas. All values are expressed as g/kgTS, with percentages respect to initial content reported in brackets.

gC/kgTS and (%)	IAa	IAb	CAa	CAb	ANa	ANb
Carbon balance						
Initial content	368 (100)	368 (100)	368 (100)	368 (100)	368 (100)	368 (100)
Final content (305 d)	181 (49.2)	232 (63.0)	208 (56.7)	303 (82.5)	282 (76.8)	274 (74.6)
Leachate emissions	7.4 (2.0)	6.9 (1.9)	9.0 (2.4)	9.1 (2.5)	9.8 (2.7)	10.3 (2.8)
Biogas emissions	52 (14)	64 (17)	44 (12)	57 (16)	28 (8)	21 (6)
Methane emissions	24 (6)	32 (9)	17 (5)	25 (7)	15 (4)	10 (3)
Missing (%)	34.74	17.73	29.06	−0.48	13.06	16.77
Nitrogen balance						
Initial content	9.70 (100)	9.70 (100)	9.70 (100)	9.70 (100)	9.70 (100)	9.70 (100)
Final content (305 d)	8.48 (88.4)	8.38 (86.4)	8.63 (89.0)	9.00 (92.8)	8.01 (82.5)	8.08 (83.3)
Leachate emissions	1.45 (14.9)	1.34 (13.8)	1.30 (13.4)	1.46 (15.1)	1.16 (12.0)	1.19 (12.3)
Nitrates emissions	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Gasseous emissions	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)
Missing (%)	−2.3	−0.2	−2.4	−7.9	5.5	4.4

4.2. Influence of pre-aeration on methane production

Acidity was the main inhibition factor for methanogenic bacteria; they work efficiently only in a small range (6.5–8.0) of pH values (Sekman et al., 2011). VFA constitute the feed for methanogens, although, if concentrations are too high (upper than 6000 mg/L), pH decreases excessively resulting in inhibition of methanogenic activity. This "excess of loading" may potentially occur in all anaerobic digesters, although can be controlled through the FOS/TAC, the ratio between VFA and alkalinity. According to Farquhar and Rovers (1973), the methane production of an anaerobic degradation system requires a ratio between VFA and alkalinity lower than 0.8. In our reactors methanogenesis started with an even higher value (1.25), but FOS/TAC decreased rapidly once methane production was established (Fig. 3).

After a pre-aeration period, total methane production in hybrid reactors was greater than that obtained from control anaerobic bioreactors, confirming that pre-aeration improves methane generation and increases the production rate.

Acidic conditions were influenced by aeration mode: intermittent aeration speeded up pH increase, shortening the first phase and anticipating methane generation (Fig. 3). The alternation of aerobic and anaerobic periods renders the transition from an aerobic to an anaerobic microbial population more dynamic, thus forming a diversified microbial community. Sang et al. (2009) carried out tests on landfill bioreactors under both anaerobic and aerobic conditions, with aeration supplied both continuously and intermittently. They found a very rich bacterial population within the reactor characterized by intermittent aeration: in particular methanogens, such as *Methanobacteriales* (H_2-utilizing groups) and *Methanosarcinales* (acetate-utilizing group) were detected. During the test *Methanobacteriales* were gradually replaced by *Methanosarcinales*, suggesting that the main degradation pathway was acetotrophy for methanogenesis. Aerobic bacteria were found to be subjected to stressful conditions due to the presence of anaerobic periods, therefore, during aerobic conditions the degradation process progressed at a faster rate.

Excessive pre-aeration should be avoided to limit the oxidation of organic substance and the subsequent decrease of methane production potential. As an example, IAa was aerated to a slightly greater extent than IAb (Fig. 2): this resulted in a faster transition to methanogenesis in IAa after the end of the aeration phase, but the total methane generation was negatively affected and a lower yield compared to IAb was observed (Fig. 6).

The Gompertz Equation proved suitable to model biogas production during the test. The results of model calibration carried out by minimizing mean square deviation of the values measured and calculated are reported in Table 5; standard deviation between real data and the model was lower than 7%. Gompertz model parameters are reported in Fig. 7 also, where they are plotted versus the total injected air.

5. Conclusions

Pre-aeration was effective in reducing the organic content of reactors, increasing degradation kinetics, controlling the concentration of VFA and stimulating methanogenesis. Moreover, air injection was found to increase total methane production in the subsequent anaerobic phase in comparison with control bioreactors where acid conditions endured for a longer time.

Intermittent aeration proved to be more efficient than continuous aeration, promoting faster reaction kinetics for methane production.

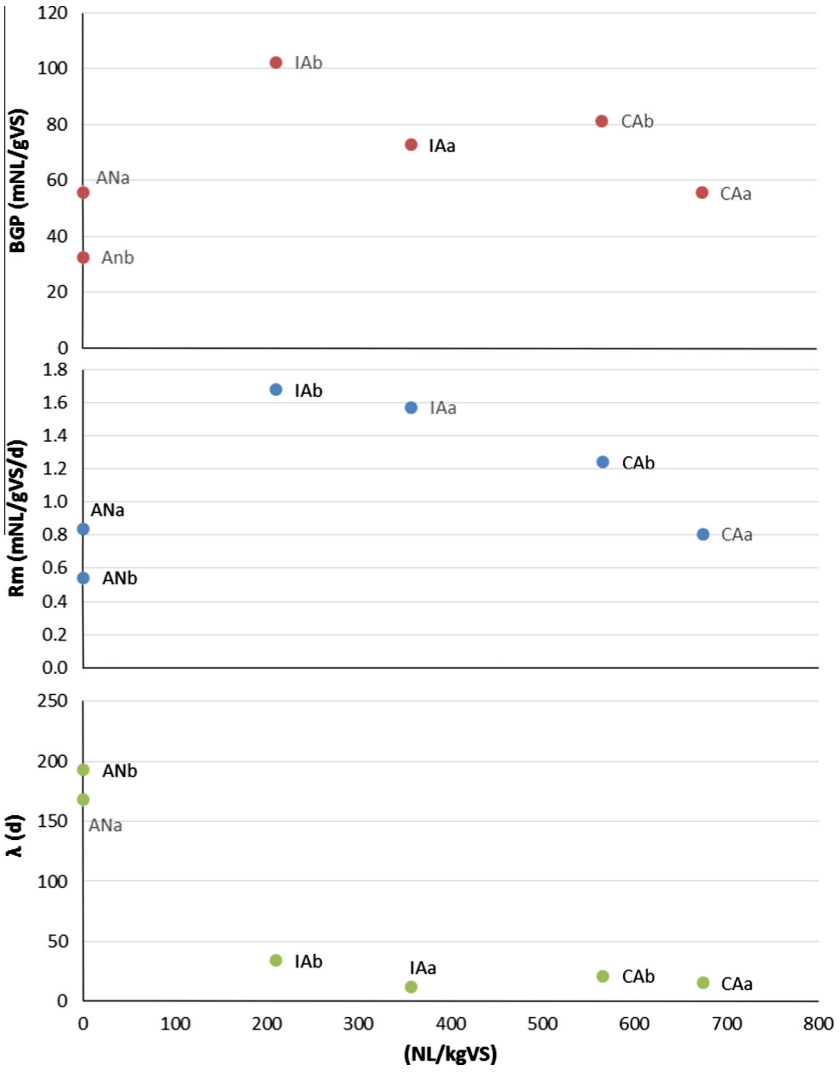


Fig. 7. Gompertz model parameters (BGP, R_m, λ) respect to total air volume injected. BGP is the methane yield potential (mL/gVS), R_m is maximal daily methane yield (mL/gVS/d), λ is bacteria growth lag time (d).

The highest methane yield was recorded in intermittent reactor IAb, where the aeration was stopped earlier (day 35) than other reactors, although the low pH value (6.25) and the still high VFA content (above 7100 mg/L) in the leachate caused a higher lag phase and a slightly slower biogas generation rate than in IAa. However, despite the better start, the longer pre-aeration in IAa caused a cumulative methane production 25% lower than in IAb.

Pre-aeration of a bioreactor landfill proved effective in promoting and speeding up methane production; the process should be carefully monitored and air injection carried out and eventually interrupted based on the biochemical performance of the reactor.

Carbon mass balance evidenced a significant difference between the analyzed and the calculated final content of TOC, probably due to waste heterogeneity rather than system losses. Carbon emissions via leachate were approximately 2–3 % of the initial TOC content in the solid waste, with no evident differences between pre-aerated and control reactors; conversely, carbon emissions via process gas were enhanced in pre-aerated reactors (12–17% of the initial TOC) compared to control ones (6–8% of the initial TOC).

Ammonia nitrogen concentration in leachate from all reactors remained above 1000 mgN/L at the end of the test; a subsequent aeration phase, as foreseen in the S.An.A. landfill concept, is required for ammonia removal and would be beneficial for the completion of waste biological stabilization.

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References

Agdag, O.N., Sponza, D.T., 2005. Effect of alkalinity on the performance of a simulated landfill bioreactor digesting organic solid wastes. Chemosphere 59, 871–879.

Christensen, T.H., Kjeldsen, P., Lindhardt, B., 1996. Gas-Generating Processes in Landfills in Landfilling of Waste: Biogas. E & FN Spon, London, ISBN 0 419 19400 2.

Chugh, S., Clarke, W., Pullammanappallil, P., Rudolph, V., 1998. Effect of recirculated leachate volume on MSW degradation. Waste Manage., 564–573

Cossu, R., Lai, T., Sandon, A., 2012. Standardization of BOD5/COD ratio as a biological stability index for MSW. Waste Manage. J. 32, 1503–1508.

Cossu, R., Raga, R., Rossetti, D., 2003. The PAF model: an integrated approach for landfill sustainability. Waste Manage. 23, 37–44.

Cossu, R., Raga, R., Vettorazzi, G., 2005. Carbon and nitrogen mass balance in some landfill models for sustainability assessment. In: Proceedings Sardinia 2005.

Tenth International Waste Management and Landfill Symposium, S. Margherita di Pula, Cagliari, Italy, 3–7 October 2005.
Cossu, R., 2009. From triangles to cycles. Waste Manage. 29, 2915–2917.
De Abreu, R., La Motta, E., McManis, K., 2005. Facultative landfill bioreactors (FLB): results of a pilot-scale study. Waste Containment Remediat., 1–13
EPA, 2015. Bioreactors, 23/04/2015, <http://www.epa.gov/solidwaste/nonhaz/municipal/landfill/bioreactors.htm#1>.
Erses, A.S., Onay, T.T., Yenigun, O., 2008. Comparison of aerobic and anaerobic degradation of municipal solid waste in bioreactor landfill. Bioresour. Technol. 99, 5418–5426.
Farquhar, G.J., Rovers, F.A., 1973. Gas production during refuse decomposition. Water, Air Soil Pollut. 2, 483–495.
Fellner, J., Doeberl, G., Allgaier, G., Brunner, P.H., 2009. Comparing field investigations with laboratory models to predict landfill leachate emissions. Waste Manage. 29, 1844–1851.
He, P., Yang, N., Gu, H., Zhang, H., Shao, L., 2011. N_2O and NH_3 emissions from a bioreactor landfill operated under limited aerobic degradation conditions. J. Environ. Sci. 23, 1011–1019.
Kjeldsen, P., Barlaz, M.A., Rooker, R., Baun, A., Ledin, A., Christensen, T.H., 2002. Present and long-term composition of MSW landfill leachate: a review. Environ. Sci. Technol. 32 (4), 297–336.
Lay, J.J., Li, Y.Y., Noike, T., 1998. Developments of bacterial population and methanogenic activity in a laboratory-scale landfill bioreactor. Water Resour. 32, 3673–3679.
Long, Y., Long, Y.-T., Liu, H.-C., Shen, D.-S., 2009. Degradation of refuse in hybrid bioreactor landfill. Biomed. Environ. Sci. 22, 303–310.
Lornage, R., Redon, E., Lagier, T., 2013. Carbon and nitrogen mass balance for different landfill concepts, Lysimeter tests. In: Cossu, R., Van der Sloot, H.A. (Eds.), Sustainable Landfilling. Cisa Publisher, pp. 95–105.
Norbu, T., Visvanathan, C., Basnayake, B., 2005. Pretreatment of municipal solid waste prior to landfilling. Waste Manage. 25, 997–1003.
Price, G.A., Barlaz, M.A., Hater, G.R., 2003. Nitrogen management in bioreactor landfills. Waste Manage. 23, 675–688.
Raga, R., Cossu, R., 2013. Bioreactor tests preliminary to landfill in situ aeration: a case study. Waste Manage. 33, 871–880.
Raga, R. et al., 2015. Landfill aeration for emission control before and during landfill mining. Waste Manage. 46, 420–429.
Repetti, R., Testolin, G., Cossu, R., Raga, R., 2013. Discarica S.AN.A®: Un sistema innovativo per la sostenibilità ambientale delle discariche. In: Proceedings Sardinia 2013. Fourteenth International Waste Management and Landfill Symposium, S. Margherita di Pula, Cagliari, Italy, 30 September–4 October 2013.
Ritzkowski, M., Heyer, K.-U., Stegmann, R., 2006. Fundamental processes and implications during in situ aeration of old landfills. Waste Manage. 26, 356–372.
Ritzkowski, M., Stegmann, R., 2013. Landfill aeration within the scope of post-closure care and its completion. Waste Manage. 33, 2074–2082.
Sandip, M., Kanchan, K., Ashok, B., 2012. Enhancement of methane production and bio-stabilisation of municipal solid waste in anaerobic bioreactor landfill. Bioresour. Technol. 110, 10–17.
Sang, N.N., Soda, S., Inoue, D., Sei, K., Ike, Michihiko, 2009. Effects of intermittent and continuous aeration on accelerative stabilization and microbial population dynamics in landfill bioreactors. J. Biosci. Bioeng. 108, 336–343.
Sekman, E., Top, S., Varank, G., Bilgili, M.S., 2011. Pilot-scale investigation of aeration rate effect on leachate characteristics in landfills. Fresenius Environ. Bull. 20.
Shao, L.-M., He, P.-J., Li, G.-J., 2008. In situ nitrogen removal from leachate bioreactor landfill with limited aeration. Waste Manage. 20, 1000–1007.
Sun, F., Wang, Y.N., Sun, X., Wu, H., Zhang, H., 2013. Production characteristics of N_2O during stabilization of municipal solid waste in an intermittent aerated semi-aerobic bioreactor landfill. Waste Manage. 33, 2729–2736.
US6398958 B1 2002. Gary R. Hater, Roger B. Green, Jeffrey M. Harris. Patent Facultative Landfill Bioreactor, Waste Management. inc.
Valencia, R., Van der Zon, W., Woelders, H., Lubberding, H.J., Gijzen, H.J., 2009. The effect of hydraulic conditions on waste stabilisation in bioreactor landfill simulators. Bioresour. Technol. J. 100, 1754–1761.
Valencia, R., Van der Zon, W., Woelders, H., Lubberding, H.J., Gijzen, H.J., 2011. Anammox: an option for ammonium removal in bioreactor landfill. Waste Manage. 31, 2287–2293.
Vigneron, V., Ponthieu, M., Barina, G., Audic, J.M., Duquennoi, C., Mazeas, L., Bernet, N., Bouchez, T., 2007. Nitrate and nitrite injection during municipal solid waste anaerobic biodegradation. Waste Manage. 27, 778–791.
Voss, E., Weichgrebe, D., Rosenwinkel, K.H., 2009. FOS/TAC-Deduction, Methods, Application and Significance, Internationale Wissenschaftskonferenz "Biogas Science 2009 – Science Meets Practice", LfL-Bayern, 2–4. 12.09, Erding.
Wang, Q., Kuninobu, M., Kakimoto, K., Ogawa, H.I., 1999. Upgrading of anaerobic digestion of waste activated sludge by ultrasonic pretreatment. Bioresour. Technol. 68, 309–313.
Wu, C., Shimaoka, T., Nakayama, H., Komiya, T., Chai, X., Hao, Y., 2014. Influence of aeration modes on leachate characteristic of landfills that adopt the aerobic–anaerobic landfill method. Waste Manage. J. 34 (2014), 101–111.
Xu, Q., Jin, Z., Ma, Z., Tao, H., Ko, J.H., 2014. Methane production in simulated hybrid bioreactor landfill. Bioresour. Technol. 168, 92–96.
Yuen, S.T.S., Styles, J.R., McMahon, T.A., 1995. An active landfill management by leachate recirculatio: a review and an outline of a full-scale project. In: Proceedings of Sardinia 95, Fifth International Waste Management and Landfill Symposium, S. Margherita di Pula, Cagliari, Italy, 2–6 October 1995. CISA, pp. 403–418.
Zehnder, A.B.J., Ingvorsen, K., Marti, T., 1982. Microbiology of methanogen bacteria in anaerobic digestion. Proceedings of the 2nd International Symposium of Anaerobic Digestion, Travemunde, 6–11 September 1981. Elsevier Biomedical Press, BV, Amsterdam, The Netherlands, pp. 45–68.